COMPUTATIONAL SOCIAL MEDIA ANALYSIS

Solidarity or Blame

How technology employment disruption is discussed across Reddit communities

Mid-term Evaluation Report, Phase 2

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Abstract

This mid-term study examines discussion of technology employment disruption in three Reddit communities. We audited 2,974,250 records and reconstructed text, threads, and replies. In a shared January-May 2025 window, layoff language appeared in 2.32% of collected r/cscareerquestions comments and 0.20% of r/technology comments, an 11.36x difference. This gives preliminary support to a community-difference hypothesis, but it is not a prevalence or causal estimate. Human coding is required before we draw conclusions about solidarity, blame, or polarization.

1 Introduction

Technology layoffs, hiring slowdowns, artificial intelligence, and restructuring are discussed in communities with different purposes and audiences. A career-focused forum may foreground first-person job-search experiences, while a general technology forum may foreground news, products, and industry-level interpretation. Comparing communities is therefore necessary before drawing conclusions about online reactions to employment disruption. This project asks how employment-disruption language varies across communities, which explanations and future expectations appear in the discussion, and how replies organize support or contestation.

The mid-term evaluation covers Phase 2 evidence: data collection, preparation, preliminary analysis, and an initial inference from a controlled comparison. The available corpus supports a within-Reddit, cross-community pilot. It does not yet support the cross-platform claim in the original synopsis. We treat the records as a collected corpus rather than a representative sample of technology workers. Missing text, unavailable parent records, and changing collection coverage remain visible in the analysis. Human coding will define the solidarity-neutral-blame labels before we evaluate any classifier.

2 Literature Review

Bose (2026), Ton (2025), and Yin, Wong, and Reddy (2025) examine different parts of employment disruption. Bose found an amplification gap on X that did not replicate on Reddit. Ton found both displacement concern and adaptation across profession-based subreddits. Yin, Wong, and Reddy found that people seek different forms of support and may receive mismatched replies. Community setting, platform design, and response function therefore belong inside the research question.

The methods literature supports a staged design. VADER is transparent but measures sentiment rather than the target of support or blame. BERT can learn contextual classes after annotation, Sentence-BERT can support sampling, and LDA or BERTopic can surface candidate themes. Network studies add a second warning: a reply or shared URL records contact, not agreement. The remaining gap is a controlled community comparison joined to human-coded solidarity, blame, and reply structure. We address it with a shared window, community-specific denominators, a 200-record coding sample, and delayed network interpretation.

2.1 Comparative Literature Survey

Paper	Year	Data and collection	Analytical method	Main result	Limitation	What this project adopts
Bose, AI layoff discourse	2026	763 X posts; 647 Reddit posts.	Keyword and account comparisons.	Capital posts led on X, not Reddit.	Preprint; narrow accounts.	Control context; audit keywords.
Yin et al., Reddit support	2025	Interviews with 16 Reddit users.	Qualitative support analysis.	Users sought several support types; replies could mismatch needs.	Small, self-selected sample.	Code support and response function.
Ton, AI as a threat	2025	72 profession subreddits, 2009-2024.	Text models, regression, synthetic control.	Threat, adaptation, and productivity coexisted.	Preprint; group-level inference.	Separate narratives; compare communities.
Naing and Udomwong, ChatGPT opinions	2024	202,905 Reddit comments, 2022-2023.	Topics, VADER, SWOT.	Job-market concerns appeared among mixed views.	Product-focused; sentiment is broad.	Use topic and sentiment baselines.
Brynjolfsson et al., Generative AI at Work	2025	5,179 support agents in one firm.	Staggered adoption analysis.	Productivity rose about 14% on average.	One firm and occupation.	Include opportunity narratives.
Eloundou et al., GPTs are GPTs	2024	O*NET tasks rated by humans and GPT-4.	Task-exposure rubric.	80% had at least 10% task exposure.	Exposure is not job loss.	Create a separate AI/work facet.
Acemoglu and Restrepo, Robots and Jobs	2020	US local markets and robot exposure, 1990-2007.	Shift-share regressions.	Robot exposure reduced local jobs and wages.	Earlier automation; ecological data.	Separate automation from other explanations.
Hutto and Gilbert, VADER	2014	Human-rated social-media corpora.	Lexicon and valence rules.	Strong, interpretable sentiment baseline.	No target, stance, or irony.	Benchmark against human labels.
Devlin et al., BERT	2019	BooksCorpus, Wikipedia, 11 NLP tasks.	Pretrain and fine-tune.	Context improved task performance.	Costly, opaque, label-dependent.	Fine-tune after codebook validation.
Reimers and Gurevych, Sentence-BERT	2019	Inference and similarity datasets.	Siamese sentence embeddings.	Enabled efficient semantic comparison.	Similarity can join opposing views.	Support sampling and theme exploration.
Blei et al., LDA	2003	Benchmark document corpora.	Probabilistic topic model.	Represented documents as topic mixtures.	Bag-of-words; topic count is subjective.	Use as an interpretable baseline.
Grootendorst, BERTopic	2022	Public benchmark text corpora.	Embeddings, UMAP, HDBSCAN, c-TF-IDF.	Produced coherent topic representations.	Sensitive to model and clustering choices.	Compare stability with LDA.
Notarmuzi et al., propagation	2022	Nearly one billion events on three platforms.	Cascade scaling analysis.	Shared diffusion patterns appeared.	No Reddit stance evidence.	Separate reach and depth from meaning.
Kristensen et al., partisan boundaries	2023	30 million posts across eight platforms.	URL-sharing networks.	Heterophily rose slightly; boundaries persisted.	Alternative-news and country context.	Do not treat edges as agreement.
Reagle, disguising Reddit sources	2022	22 studies; 10 author interviews.	Search-based ethics audit.	Quoted Reddit text often remained discoverable.	Small sample; tools change.	Exclude record-level text and usernames.

Table 1. Fifteen studies reviewed for substantive framing, measurement, network interpretation, and research ethics.

3 Project Objectives

- Collect and consolidate employment-disruption discussion from multiple Reddit communities using a documented, reproducible workflow.
- Prepare text, author, timestamp, thread, and reply fields while retaining quality indicators and missing-context states.
- Compare layoff, hiring, AI/work-change, restructuring, economic/policy, and career-transition language across communities in a shared observation window.
- Develop validated solidarity-neutral-blame stance and thematic labels through human coding, then connect them to reply networks and time trends.

3.1 Research Hypotheses and Mid-term Status

Hypothesis	Test at this stage	Status
H1 Community concentration	Employment-disruption language differs between r/cscareerquestions and r/technology in the same observation window.	Preliminary support
H2 Discourse function	Career discussion contains more worker-centred support and experience narratives; technology discussion contains more news and industry framing.	Not yet tested
H3 Reply structure	Validated solidarity and blame labels are associated with different reply and engagement patterns.	Not yet tested

Table 2. Hypotheses are separated from current evidence to avoid treating planned analyses as findings.

4 Methodology

4.1 Data collection

The available Reddit corpus contains 115,213 posts and 2,859,037 comments stored in 29,821 JSON batches across 21 monthly folders from January 2025 to September 2026. Communities are r/cscareerquestions, r/programming, and r/technology. Collection settings and completeness are unknown, so the analysis reports observed records and coverage rather than community-wide activity.

4.2 Data preprocessing

Raw JSON was retained unchanged. The team normalized whitespace and HTML entities, separated line-prefixed quoted text from own text, created stable pseudonymous author keys for known usernames, and reconstructed structural threads and immediate reply edges. Removed and deleted text states remain explicit. The prepared dataset contains 2,974,250 records, 164,271 structural threads, and 2,859,037 reply edges. Comment body text is available for 94.61% of comments; 65.96% link to a stored submission.

4.3 Retrieval and comparative analysis

Versioned lexical screens retrieve six employment-disruption facets: layoffs, hiring conditions, AI/work change, corporate restructuring, economic or policy discussion, and career transitions. URLs and line-prefixed quotations are excluded from the screen. Facet flags are candidate retrieval aids, not validated relevance or causality. To address the instructor feedback on comparison, the primary comparison uses comments from r/cscareerquestions and r/technology in the shared January-May 2025 window. Rates use all collected comments in each community-window as the denominator.

4.4 Qualitative and network preparation

A deterministic 200-record reading sample balances community, quarter, record type, score band, and seven sampling groups. Blank coding fields cover relevance, document type, experiences, explanations, future expectations, themes, emotions, response function, and notes. Gephi files include an author reply network and a conversation reply network. Human coding and Gephi interpretation are next-phase work.

4.5 Analysis and validation plan

Stage	Input	Operation	Output and validation
Corpus QA	Raw Reddit JSON batches	Check dates, IDs, text states, duplicates, and parent availability.	Coverage manifest, quality rates, and reproducible prepared tables.
Candidate retrieval	Own text after quote and URL removal	Apply versioned screens for six employment-disruption facets.	Candidate flags only; inspect false positives during coding.
Controlled comparison	Comments from two subreddits, January-May 2025	Calculate within-community rates, rate ratios, and a two-proportion z-test.	H1 evidence with denominators and independence caveat.
Human coding	Balanced 200-record reading sample with thread context	Two coders refine the codebook and double-code an overlap.	Relevance, stance, target, theme, and response labels plus agreement.
NLP evaluation	Human-coded records	Compare lexical, VADER, and transformer baselines.	Report macro F1, per-class precision/recall, confusion matrix, and errors.
Network analysis	Reply graphs joined to validated labels	Measure components, degree, reciprocity, and conversation depth by label.	Interpret structure only alongside coded text; no degree-as-influence claim.

Table 3. Methods are ordered so that retrieval, validation, comparison, and interpretation remain separate.

For a facet f in community c, the descriptive rate is n(f,c) divided by all collected comments in c during the shared window. The rate ratio compares the two community rates. The z-test pools the two proportions under a null of equal rates. Because comments are clustered within authors and threads, its p-value is not treated as a population-level causal test. The planned stance model will use grouped or time-aware splits where possible so that near-duplicate comments and repeated authors do not leak across training and evaluation.

5 Results and Discussion

5.1 Corpus and quality profile

Measure	Result
Posts	115,213
Comments	2,859,037
Total records	2,974,250
Comment body available	94.61%
Structural threads	164,271
Prepared reply edges	2,859,037
Direct layoff comments	28,077
Broader facet candidates	177,808 distinct records

Table 4. Audited corpus profile from the team archive.

5.2 Controlled community comparison

Facet	r/ cscareerquestions	Rate	r/technology	Rate	Ratio
Layoffs	5958	2.3228%	3260	0.2045%	11.36
Hiring conditions	21794	8.4968%	7244	0.4545%	18.69
AI and work change	7585	2.9571%	8725	0.5474%	5.4
Restructuring	2652	1.0339%	1377	0.0864%	11.97
Economic or policy	3516	1.3708%	3918	0.2458%	5.58
Career transitions	3240	1.2632%	1998	0.1254%	10.07

Table 5. Facet counts and rates in collected comments, January-May 2025. Facets overlap.

The comparison gives a clear preliminary inference: employment-disruption language occupies a larger share of the collected r/cscareerquestions comment stream than of r/technology in the shared window. Layoff language appears in 2.32% of career-question comments and 0.20% of technology comments, an 11.36x rate ratio. Hiring-condition language shows the largest contrast at 8.50% compared with 0.45% (18.69x). AI/work-change language occurs in both communities, with rates of 2.96% and 0.55% (5.40x). The layoff-rate difference gives $z = 141.42$ and $p < 0.001$ under an independence

assumption. Repeated authors violate that assumption, so the test is descriptive evidence for H1 rather than a population estimate. It does not correct selection bias, changing coverage, or lexical false positives.

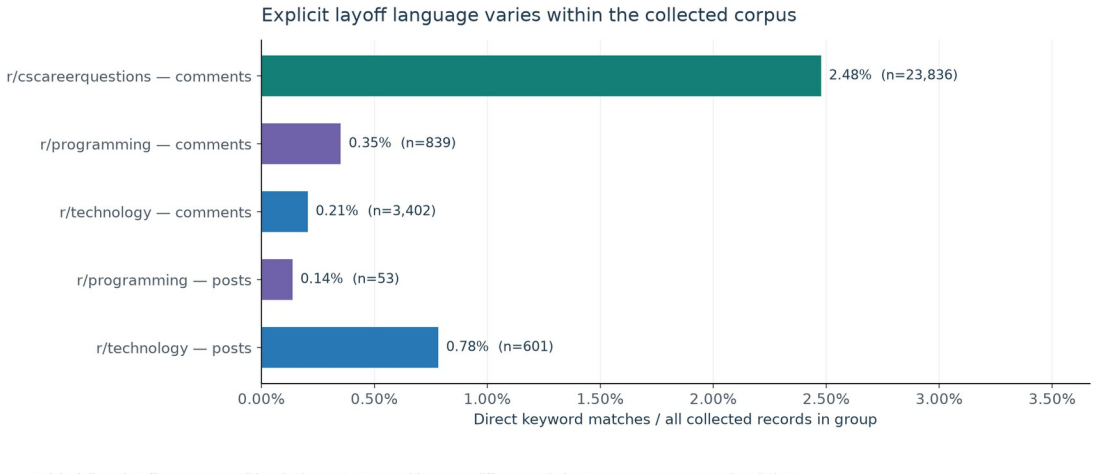


Figure 1. Direct layoff-language shares by community. The plot is descriptive and reflects collected coverage.

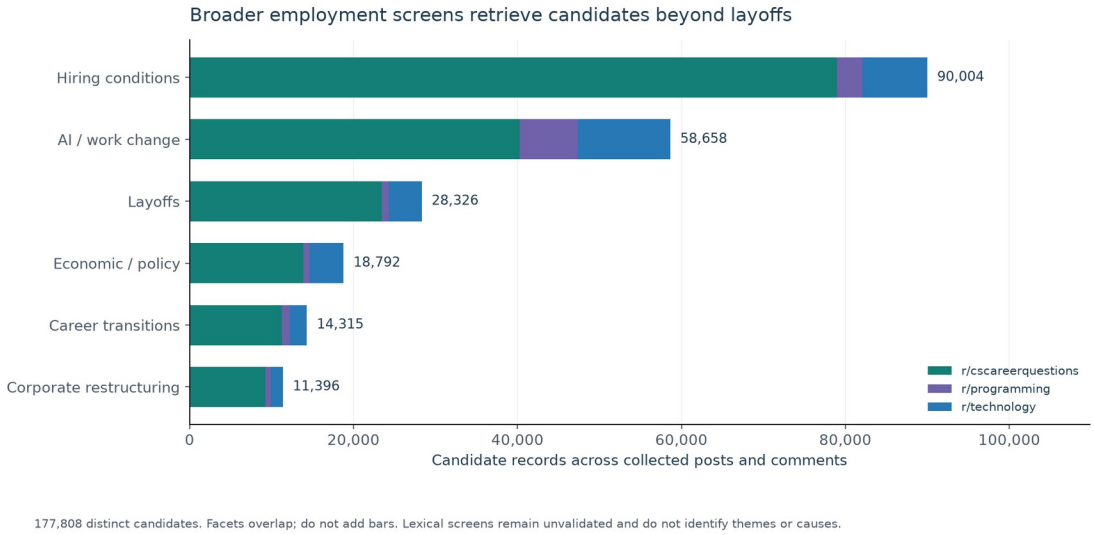


Figure 2. Broader employment-disruption candidate counts. Facets overlap and require human review.

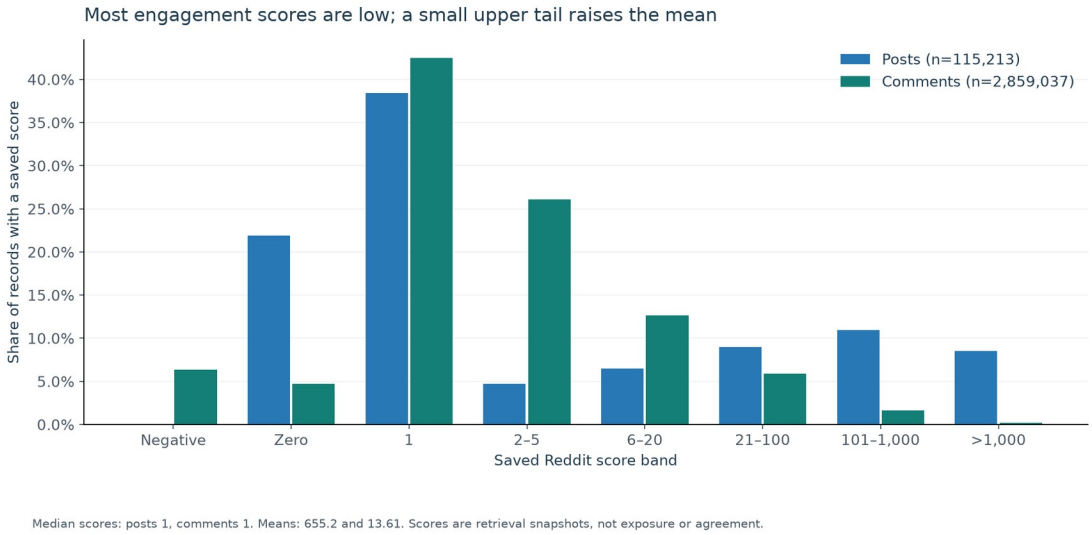


Figure 3. Saved-score distribution. Median scores are 1 for both posts and comments, so means alone are misleading.

5.3 Exploratory stance screen

Exploratory label	Records	Share
blame-leaning	35	17.5
mixed	16	8.0
no lexical signal	118	59.0
solidarity-leaning	31	15.5

Table 6. Transparent lexical screen over the 200-record reading sample.

The exploratory screen finds 31 solidarity-leaning, 35 blame-leaning, 16 mixed, and 118 records with no lexical signal. The high no-signal count is expected because a small lexicon cannot identify all forms of empathy, anger, irony, or disagreement. The result is useful as a coding and model-development baseline only. It is not a sentiment estimate, and it does not establish that a post author experienced a layoff or blamed a particular actor.

5.4 Current inference and next phase

At mid-term, H1 has preliminary support: direct and indirect employment-disruption language is more concentrated in the collected career-question discussion than in general technology discussion during the same period. This does not show that one community is more negative. Different purposes, audiences, and collection composition could all contribute to the gap. H2 and H3 remain open. The team will manually code the 200-record sample, measure agreement on a shared subset, evaluate a stance classifier against those labels, analyze themes with LDA or BERTopic, and then interpret the Gephi networks alongside coded text. Claims about solidarity, blame, or polarization will require those validated labels and reported denominators.

6 Limitations and Ethical Considerations

- Collection settings, query boundaries, and API completeness are unknown.
- Comment coverage ends at different dates by community, so pooled time trends can confound changing coverage with changing discussion.
- Lexical screens can include false positives and miss indirect or non-English discussion; facet counts overlap.
- No participant occupation, location, employment status, or identity is inferred from subreddit membership or username.
- Pseudonymous author keys support structural analysis but do not guarantee anonymity. Results describe public online discourse and should not be generalized to all technology workers.

7 References

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8 Contribution of Each Team Member

Name	Role	Contribution for mid-term
Vanshit Ahuja	Research design and project lead	Audited the teammate archive and outputs; reframed the study as a within-Reddit pilot; designed the shared-window comparison and H1-H3; interpreted the statistics and limitations; researched and revised the literature review, report, and presentation; completed privacy checks, testing, repository cleanup, public release, and Phase 2 coordination.
Vaibhav Jain	Data collection lead	Set up and completed the Reddit data collection, maintained the raw batches, recorded source provenance, and checked collection coverage.
Dividh Vijay	Network and Gephi lead	Prepared the author-reply and conversation networks, exported Gephi node and edge files, checked graph direction and weights, and documented network scope, exclusions, and planned analysis.
Tanya	Data processing and quantitative analysis	Cleaned and structured the corpus, handled missing and deleted text states, linked records to threads, ran data-quality checks, and produced descriptive statistics, community comparisons, and figures.
Krish	NLP and coding lead	Developed the lexical screens and exploratory stance baseline, prepared the reading sample and coding framework, and supported the NLP evaluation plan and presentation preparation.